PROJECT REVIEW & USER STUDY FEEDBACK REPORT

Medical Hallucination Detection in Large Language Models Using Hidden-State Probing

Project Group: Group 76 (G076)	**Department:** Computer Science & Engineering
Faculty Guide: Dr. Swapna T R (Associate Professor)	**Evaluation Cohort:** N = 35 Verified Evaluators
Platform Tested: https://sonu-2006.github.io/finalhidden/	**Survey Format:** Standardized 10-Item Instrument

1. User Evaluation Study Plan & Methodology

1.1 Objective: To gather structured academic and field review feedback on the RTHD research platform. Evaluators interacted with the live deployment (https://sonu-2006.github.io/finalhidden/), inspected hidden-state vectors, tested diagnostic prompt cases in the Detection Playground, and evaluated real-time early intervention dynamics.
1.2 Participant Demographics (N = 35): Students (42.9%, n=15), Working Professionals in AI/Software (31.4%, n=11), Faculty/Medical Researchers (20.0%, n=7), and Healthcare Informaticians (5.7%, n=2).
1.3 Testing Protocol: Participants evaluated platform usability, 4096-d vector interpretability, 2D PCA transitions, lead-time utility (14-32 tokens), and hallucination suppression completeness.

2. Questionnaire Battery & Quantitative Response Analysis (N = 35)

Q#	Question & Evaluated Platform Dimension	Type / Scale	Mean (1-5)	Pos. Rating (4-5)
Q1	**Role:** What is your role? (Student, Faculty, Working professional, Other)	Demographic	N/A	100% Valid
Q2	**System Usefulness:** How useful did you find the real-time hallucination detection & probe monitoring platform? (1 = Not useful, 5 = Extremely useful)	1 to 5 Scale	**4.74**	97.1% (34/35)
Q3	**Lead-Time Importance:** How important is early detection of medical hallucinations (14–32 tokens ahead) compared to waiting for full response generation? (1 = Not important, 5 = Extremely important)	1 to 5 Scale	**4.80**	97.1% (34/35)
Q4	**Hidden-State Explorer:** How useful did you find the Hidden-State Explorer (tensor shapes, 2D PCA projection, and activation heatmap)? (1 = Not useful, 5 = Extremely useful)	1 to 5 Scale	**4.63**	91.4% (32/35)
Q5	**Detection Playground:** How useful did you find the Detection Playground feature for testing clinical prompts with adjustable risk thresholds? (1 = Not useful, 5 = Extremely useful)	1 to 5 Scale	**4.69**	94.3% (33/35)
Q6	**Explainability:** How important is it for the system to explain internal representation shifts using 4096-d vector channels and layer norms? (1 = Not important, 5 = Extremely important)	1 to 5 Scale	**4.51**	88.6% (31/35)
Q7	**Valuable Features:** Which features did you find most valuable? (Multi-select: Playground 88.6%, PCA Projection 85.7%, Trajectory Analysis 80.0%, Heatmap 77.1%, Vector Inspector 71.4%)	Multi-Select	N/A	High Engagement
Q8	**Deployment Concerns:** What concerns did you have about clinical deployment? (False alarms interrupting safe advice 42.9%, Need for larger 70B+ model probes 34.3%, Hardware requirements 28.6%, No major concerns 40.0%)	Multi-Select	N/A	Critical Feedback
Q9	**Recommendation:** How likely are you to use or recommend this platform for medical LLM auditing and safety research? (1 = Very unlikely, 5 = Very likely)	1 to 5 Scale	**4.77**	97.1% (34/35)
Q10	**Qualitative Feedback:** What additional features or improvements would you suggest based on your experience exploring the website?	Open Text	N/A	100% Response

3. Summary of Qualitative Feedback (Q10 Thematic Synthesis)

Feedback Theme	Representative User Quotes & Actionable Observations
Visual Interpretability	*'The 2D PCA cluster shift and the token risk timeline make the internal model uncertainty very tangible. You can see the probe detect the issue before the text turns dangerous.'*

Pre-Emptive Intervention	*'Truncating generation while maintaining 85-95% response completeness is practical. It delivers the factual premise without letting the model fabricate contraindications.'*
Hospital EHR Integration	*'Recommend building an EHR plugin/API wrapper so clinicians can see real-time risk banners inside medical record streaming workflows.'*

4. Key Conclusions & Verification Sign-Off

The user study validates the scientific merit and practical usability of the RTHD platform with an overall approval rating of **94.6%** across all evaluation dimensions. Evaluators confirmed that real-time token probing delivers tangible latency and safety advantages over traditional post-hoc LLM evaluation.

Submitted By: Group 76 (Dept. of CSE)	**Project Guide:** Dr. Swapna T R (Associate Professor)
Deliverable: User Study / Field Study PDF	**Status:** Formally Verified & Ready for Capstone Evaluation